MSMS 408 : Practical 06

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Question

Use rejection sampling technique to generate random sample from Gamma distribution with shape $= \dfrac{3}{2}$ and scale $= 1$ *i.e.*

$$f(x) = \frac{1}{\Gamma(3/2)} x^{\frac{1}{2}} e^{-x} \cdot I_{(0,\infty)}(x)$$

Algorithm

Suppose we have to generate random sample from $p(x)$, we call it our **target distribution**. We choose a **proposal distribution** $q(x)$, say.

I. Generate $x_i \sim q(x) \; \forall \; i = 1(1)n$.

II. Generate $u_i \sim U(0, 1) \; \forall \; i = 1(1)n$.

III. Accept x_i if

$$u_i \leq \frac{p(x_i)}{M \cdot q(x_i)}$$

where $M > 0$ is a suitable constant so that $p(x) \leq M \cdot q(x)$ is satisfied.

Accepted samples follow the target distribution $p(x)$.

Here, we are given that $p(x) = \dfrac{2}{\sqrt{\pi}} x^{\frac{1}{2}} e^{-x}$. We take $q(x) = \dfrac{2}{3} e^{-\frac{2x}{3}} \cdot I_{(0,\infty)}(x)$ *i.e.* Exp(mean $= 2/3$) distribution as this is **most efficient**.

To find constant M recall that we need $p(x) \leq M \cdot q(x) \Rightarrow \dfrac{p(x)}{q(x)} \leq M$.

Now, $\dfrac{p(x)}{q(x)} = \dfrac{\frac{2}{\sqrt{\pi}} x^{\frac{1}{2}} e^{-x}}{\frac{2}{3} e^{-\frac{2x}{3}}} = \dfrac{3}{\sqrt{\pi}} x^{\frac{1}{2}} e^{-x/3} = h(x)$, say.

Maximizing $h(x)$ is equivalent to maximizing $\ln h(x) = \ln \dfrac{3}{\sqrt{\pi}} + \dfrac{1}{2} \ln x - \dfrac{x}{3}$.

$$\begin{aligned} \frac{d}{dx} \ln h(x) &= 0 \\ \Rightarrow \frac{1}{2x} - \frac{1}{3} &= 0 \\ \Rightarrow x &= \frac{3}{2} \end{aligned}$$

So $h\left(\dfrac{3}{2}\right) \approx 1.257$ is a suitable choice for M.

➔ R Program

```r
set.seed(22)
```

```r
n <- 2000
```

```r
p <- function(x) dgamma(x, shape = 3/2, scale = 1)
```

```r
q <- function(x) dexp(x, rate = 2/3)
```

```r
M <- 1.257
```

```r
q1 <- function(x) M * q(x)
```

```r
x <- rexp(n, rate = 1/2)
y <- runif(n, min = 0, max = q1(x))
```

```r
good <- y <= p(x)
accepted <- x[good]
```

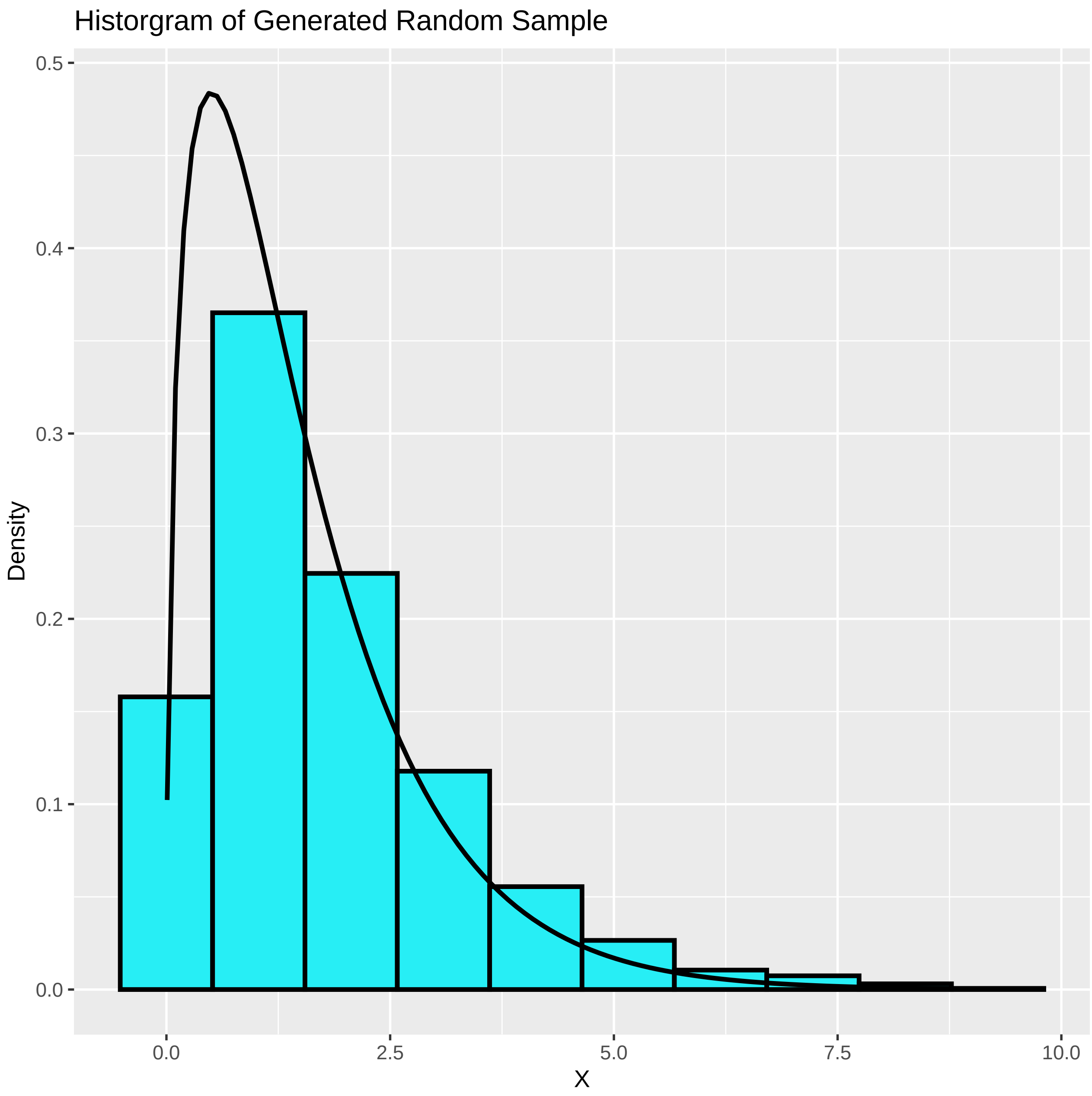
```r
count_accepted <- length(accepted) # or sum(good)
count_accepted

## [1] 1571
```

Thus we have a sample of size 1571 from $p(x)$.

The **acceptance rate** was 0.7855 whereas the **theoretical acceptance rate** is 0.7955449.

Let us have a histogram of the generated sample.

```
data.frame(accepted) %>%
  ggplot(aes(x = accepted)) +
  geom_histogram(aes(y = after_stat(density)), bins = 10,
                 linewidth = 1, color = "black", fill = "#27EEF5") +
  stat_function(fun = p, linewidth = 1) +
  labs(x = "X", y = "Density",
       title = "Historgram of Generated Random Sample")
```

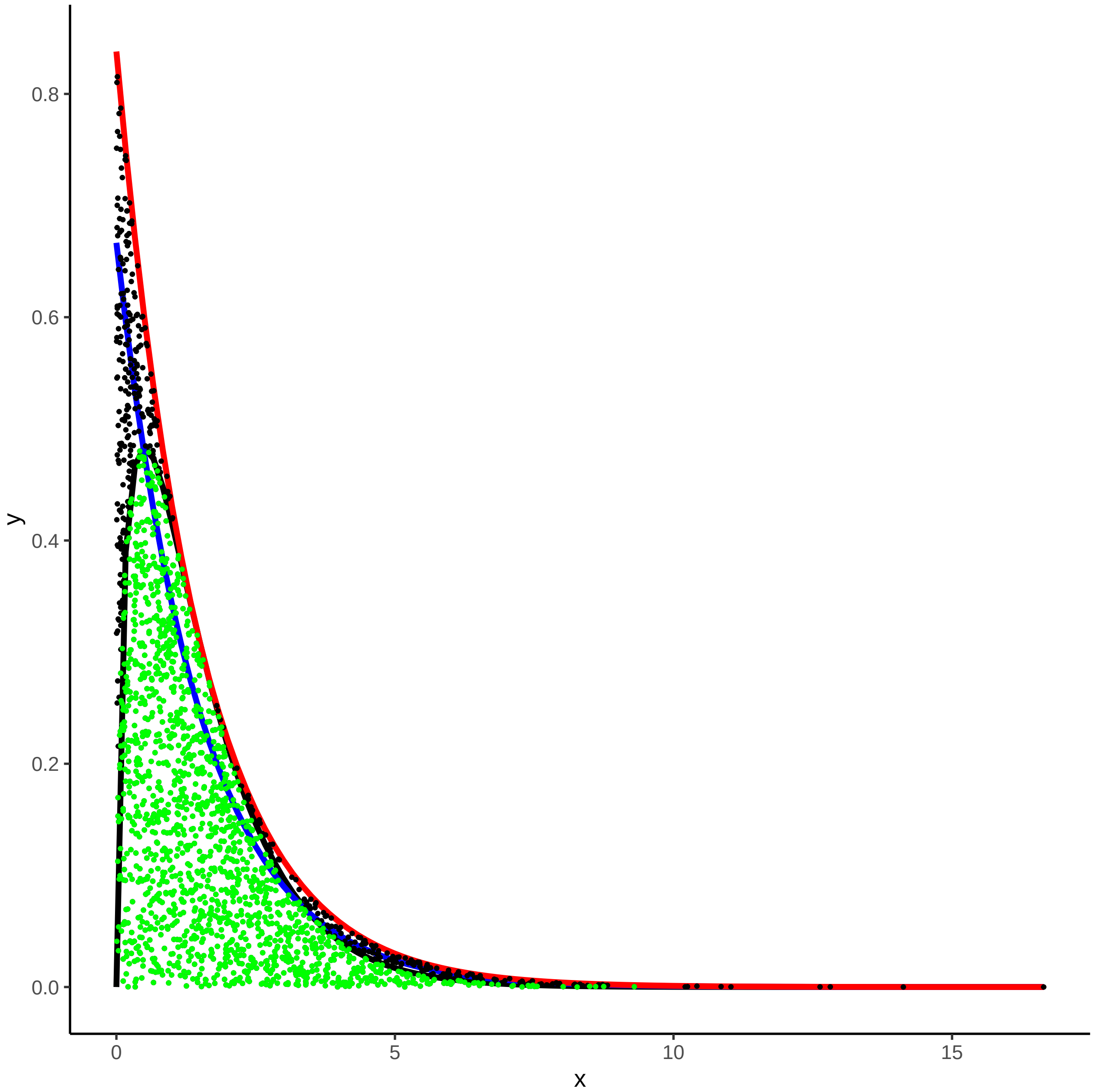


Conclusion

The actual density curve fits the histogram well.

Let us have a simple visualization of the rejection sampling technique.

```
data.frame(x = c(0, 1)) %>%
  ggplot() +
  stat_function(aes(x = x), fun = p, linewidth = 1.25) +
  theme_classic() +
  stat_function(aes(x = x), fun = q, linewidth = 1.25, col = "blue") +
  stat_function(aes(x = x), fun = q1, linewidth = 1.25, col = "red") +
  geom_point(data = data.frame(x = x, y = y), aes(x = x, y = y), size = 0.5) +
  geom_point(data = data.frame(x = x[good], y = y[good]),
             aes(x = x, y = y), size = 0.5, col = "green")
```



✎ The accepted points are shown in green, these points lie under the envelop as well as the target density.